1	2	3	4	5	6	Total
/20	/20	/20	/20	/20	/20	/100

Name: ______________________ November 11, 2018 Sunday

Department/School:__________________ SID:______________________

University Physics II

Midterm Examination

Kuang Yaming Honors School, Nanjing University

<u>Select five out of following six problems.</u>

1. (20pts.) The electric potential of a spherically symmetric charge distribution is given by

$$V(\boldsymbol{r}) = \frac{q}{4\pi\varepsilon_0}\frac{e^{-\alpha r}}{r}$$

where ε_0 is the dielectric constant of the vacuum, $r = |\boldsymbol{r}|$ and $\boldsymbol{r}$ is the position vector, and q and α are constants.

(a) Find the corresponding electric field $\boldsymbol{E}(\boldsymbol{r})$ everywhere except the point $r = 0$.

(b) What is the electric charge enclosed in a sphere of radius R centered at $r = 0$? What is the total charge of the charge distribution?

(c) Find the electric charge density $\rho(\boldsymbol{r})$ everywhere except the point $r = 0$.

(d) There is a point charge at the center of the distribution. Find the charge quantity of the point charge.

(Hint: In the spherical coordinates, $\mathrm{grad}(\varphi) = \frac{\partial\varphi}{\partial r}\boldsymbol{e}_r + \frac{1}{r}\frac{\partial\varphi}{\partial\theta}\boldsymbol{e}_\theta + \frac{1}{r\sin\theta}\frac{\partial\varphi}{\partial\phi}\boldsymbol{e}_\phi$;

$\mathrm{div}(\boldsymbol{f}) = \frac{1}{r^2}\frac{\partial(r^2 f_r)}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial(\sin\theta f_\theta)}{\partial\theta} + \frac{1}{r\sin\theta}\frac{\partial f_\phi}{\partial\phi}$, where $\boldsymbol{f} = f_r\boldsymbol{e}_r + f_\theta\boldsymbol{e}_\theta + f_\phi\boldsymbol{e}_\phi$.)

2.(20pts.) There is an infinite sheet of uniform areal charge density σ located at the plane $z = 0$ in the space. The upper half space $z > 0$ is filled with dielectrics of permittivity (or dielectric constant) ε_1 and the lower half space $z < 0$ is filled with dielectrics of permittivity ε_2.

(a) Find the electric displacement $\boldsymbol{D}$ in both upper half ($z > 0$) and lower half ($z < 0$) spaces.

(b) What is the electric field $\boldsymbol{E}$ in both upper half ($z > 0$) and lower half ($z < 0$) spaces?

3. (20pts.) There are two concentric conducting spherical shells of radii r_1 and r_2 ($r_1 < r_2$) respectively. The inner shell of radius r_1 carries an electric charge q and the outer shell of radius r_2 carries an electric charge $-q$.

(a) What is the self-energy U_1 of the electric charge distributed on the inner shell only?

(b) What is the self-energy U_2 of the electric charge distributed on the outer shell only?

(c) What is the interaction energy between the charges on the two shells? What is the total electric potential energy of the system?

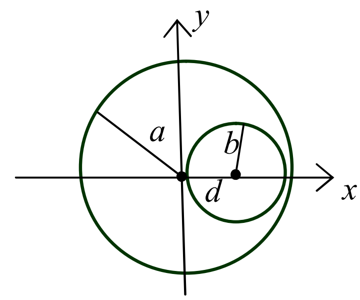
(d) Find the electric capacitance of the configuration of two conducting shells through the expression of the total electric potential energy obtained in (c).

(Hint: Here the self-energy is the potential energy of a charge distribution in the electric potential produced by the charge distribution itself. The energy is actually located in the electric field produced by the electric charge distribution and can be calculated by integral of the energy density of the electric field.)

4. (20pts.) An infinite long cylindrical conductor of radius a whose axis is coincident with the z-axis carries an electric current I that flows in the positive z-direction and its current density is uniform across the cross-section.

(a) Show that the magnetic induction $\boldsymbol{B}$ at cylindrical polar coordinates (ρ, ϕ, z) within the conductor $(\rho < a)$ can be written as

$$\boldsymbol{B} = \frac{\mu_0}{2} \frac{I\rho}{\pi a^2} \boldsymbol{e}_\phi = \frac{\mu_0}{2} \frac{I}{\pi a^2} \boldsymbol{e}_z \times \boldsymbol{\rho}$$



where $\boldsymbol{\rho} = \rho \boldsymbol{e}_\rho$, and $\boldsymbol{e}_\rho$, $\boldsymbol{e}_\phi$ and $\boldsymbol{e}_z$ are 3 orthogonal unit vectors (bases) of cylindrical coordinates.

(b)There is an infinite long cylindrical cavity of radius b $(b < a)$ in the cylindrical conductor and the axis of the cavity that is parallel to the z-axis is located at $\boldsymbol{\rho} = d\boldsymbol{e}_x$. Given that the electric current density across the cross-section with a hole is just the same as the electric current density across the hole-less cross-section in (a), show that the magnetic induction within the cavity is

$$\boldsymbol{B} = \frac{\mu_0}{2} \frac{Id}{\pi a^2} \boldsymbol{e}_y .$$

5. (20pts.) Suppose, according to Dirac's hypothesis, there are two magnetic monopoles with magnetic charges g and $2g$ ($g > 0$) that are distributed at $x = -a$ and $x = a$ on the x-axis, respectively. A magnetic dipole of moment $\boldsymbol{m} = m\boldsymbol{e}_x$ is placed at the origin $\boldsymbol{r} = 0$.

(a) What is the magnetic field $\boldsymbol{B}(x, y, z)$ produced by the two magnetic monopoles?

(b) What is potential energy of the dipole in the field of the two monopoles?

(c) What is the force $\boldsymbol{F}$ acted on the dipole?

6. (20pts.) A monochrome plane electromagnetic wave in free space (i.e., in the vacuum) can be expressed as

$$\boldsymbol{E} = \boldsymbol{E}_0 e^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)}, \quad \boldsymbol{B} = \boldsymbol{B}_0 e^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)}$$

where $\omega = c|\boldsymbol{k}| = ck$ and c is the speed of light in the vacuum.

(a) Show, by applying Maxwell equations, that

$$\boldsymbol{E} = -\frac{c^2}{\omega}\boldsymbol{k}\times\boldsymbol{B}.$$

(b) In free space, there are two travelling electromagnetic waves whose magnetic fields are

$$\boldsymbol{B}_1 = Ae_y e^{i(\boldsymbol{k}_1\cdot\boldsymbol{r}-\omega t)}, \quad \boldsymbol{B}_2 = Ae_y e^{i(\boldsymbol{k}_2\cdot\boldsymbol{r}-\omega t)}$$

where A is a constant, $\boldsymbol{k}_1 = k(\sin\theta\boldsymbol{e}_x + \cos\theta\boldsymbol{e}_z)$ and $\boldsymbol{k}_2 = k(-\sin\theta\boldsymbol{e}_x + \cos\theta\boldsymbol{e}_z)$. What are the corresponding electric fields? What is the Poynting vector (energy current density) of the resultant wave?

(c) What is the energy passing through the plane $z = 0$ per unit area per unit time?